

Neural Network Zoo

By: Miguelangel Sanchez-Yevenes

Introduction to Neural Networks

Neural networks are a key part of Artificial Intelligence. They are inspired by the human brain hence the name Neural. It's designed to recognize patterns, learn from the data then make the decisions on detection.

Layers

Three layers are shown during the neural network nodes, Input layer, Hidden Layer and Output layer. The Input receives the data, the Hidden processes the information and then the Output produces the results.

When many neurons are connected together, they form a network capable of solving complex problems.

The Zoo Concept

The Neural Network Zoo, created by The Asimov Institute, is a popular visual cheat sheet that categorizes and explains various neural network architectures by representing them as diagrams with distinct node structures.

Visual Chart

Name	Represents	What is Known for	Application Example
(CNN) Cheetah	Convolutional Neural Network	Speed and accuracy in image processing	Image recognition
(RNN) Raccoon	Recurrent Neural Network	Remembering previous information	speech recognition
(LSTM) Lemur	Long Short-Term Memory	Strong long-term memory handling	time-series prediction

Animal I chose: CNN (Cheetah)

CNNs break images into small parts and look for patterns like edges and shapes. Then they combine those patterns to recognize objects.

CNNs use the same weights across different parts of an image, allowing them to detect features regardless of where they are located which is called Parameter Sharing.